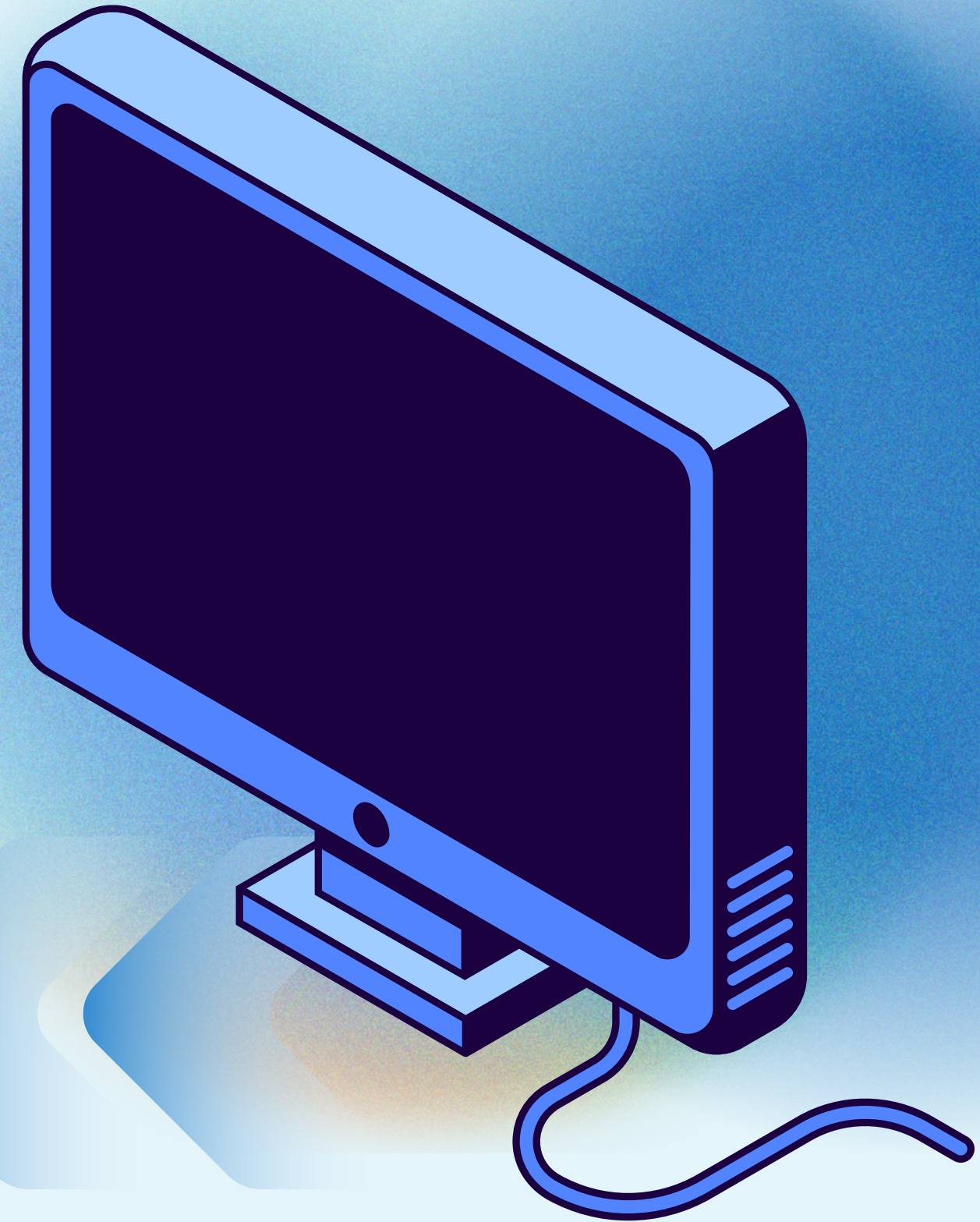


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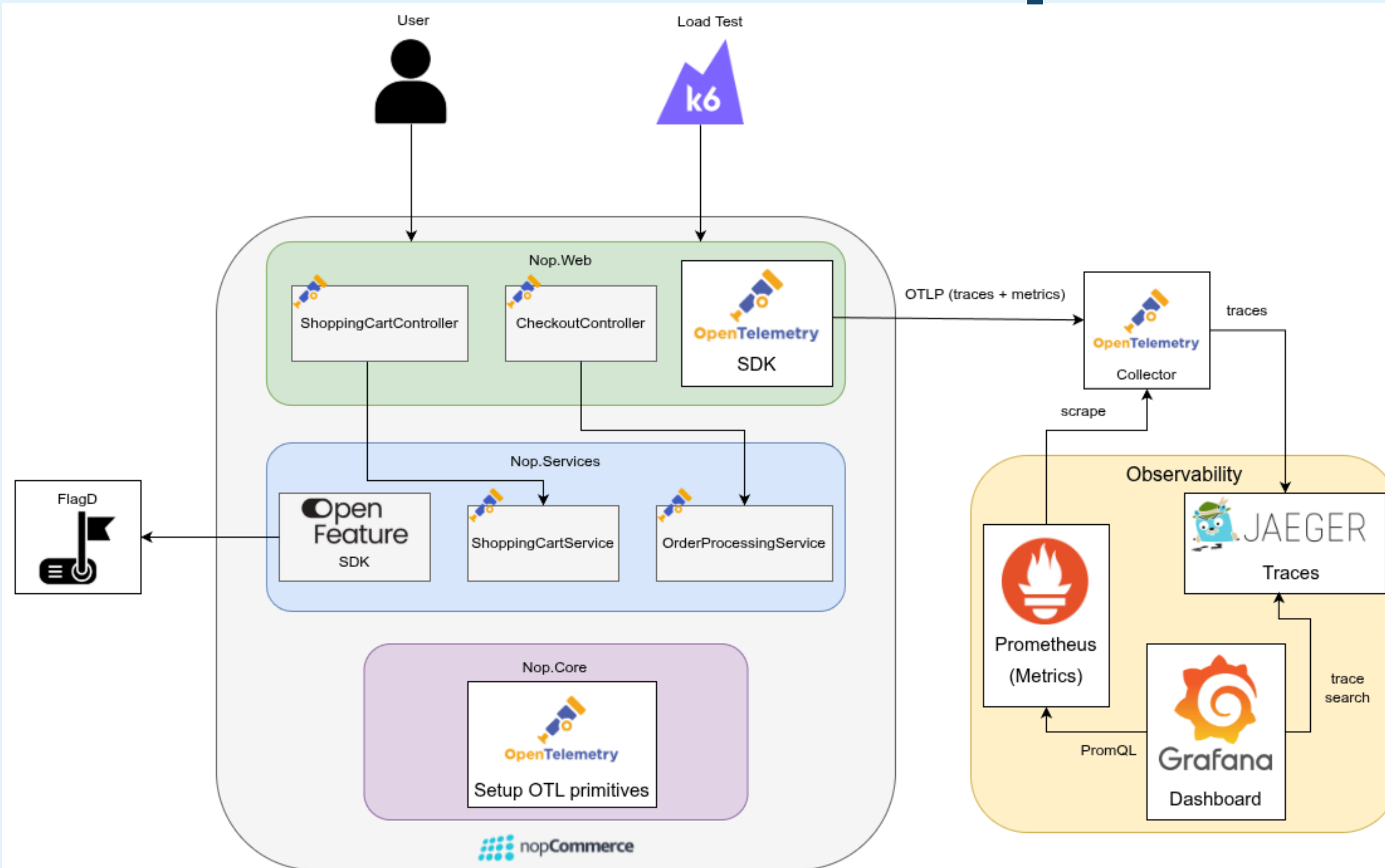
Assignment 1

March 25, 2026

Rodrigo Abreu, 113626



Architecture – Customer places order



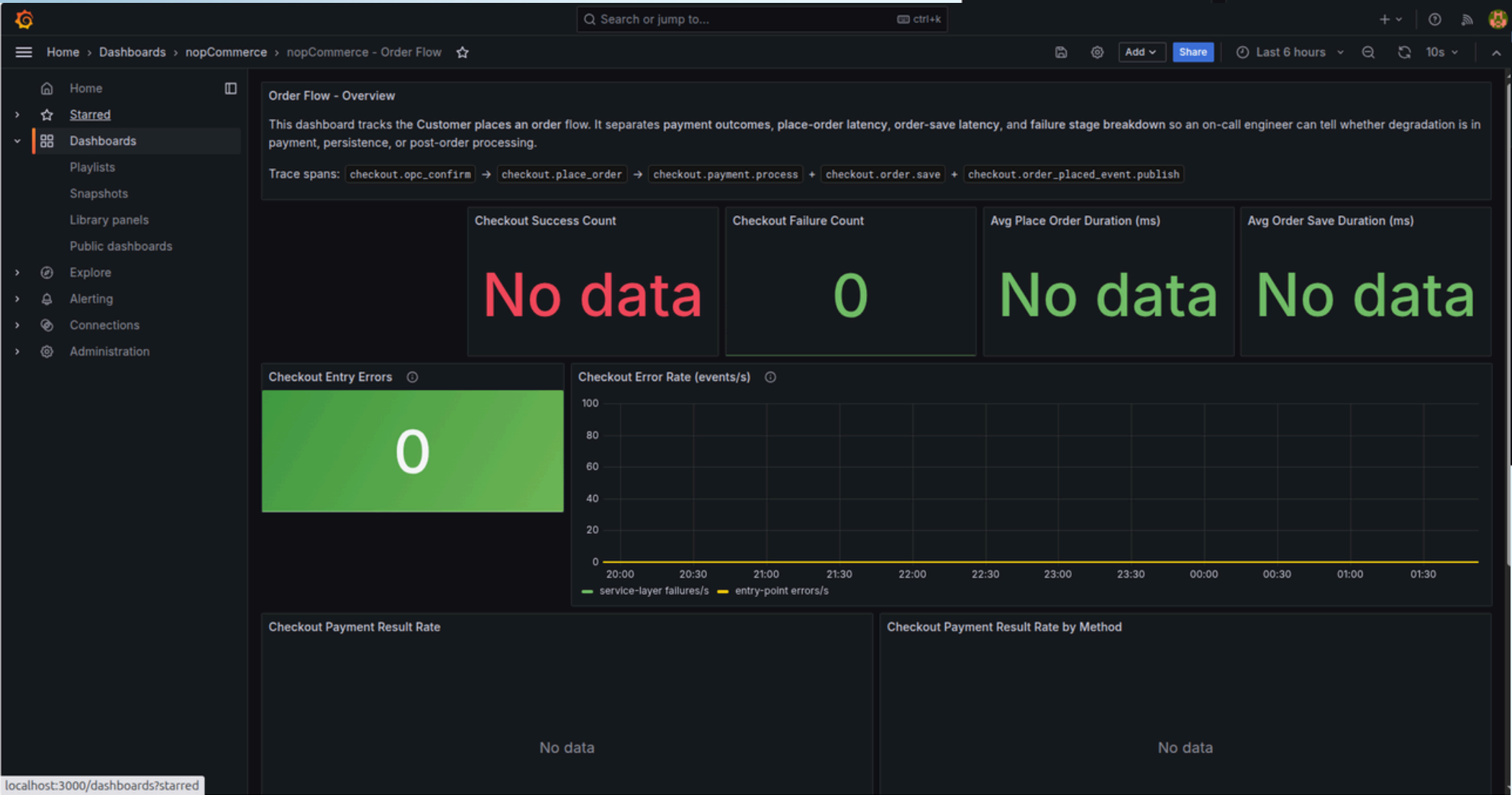
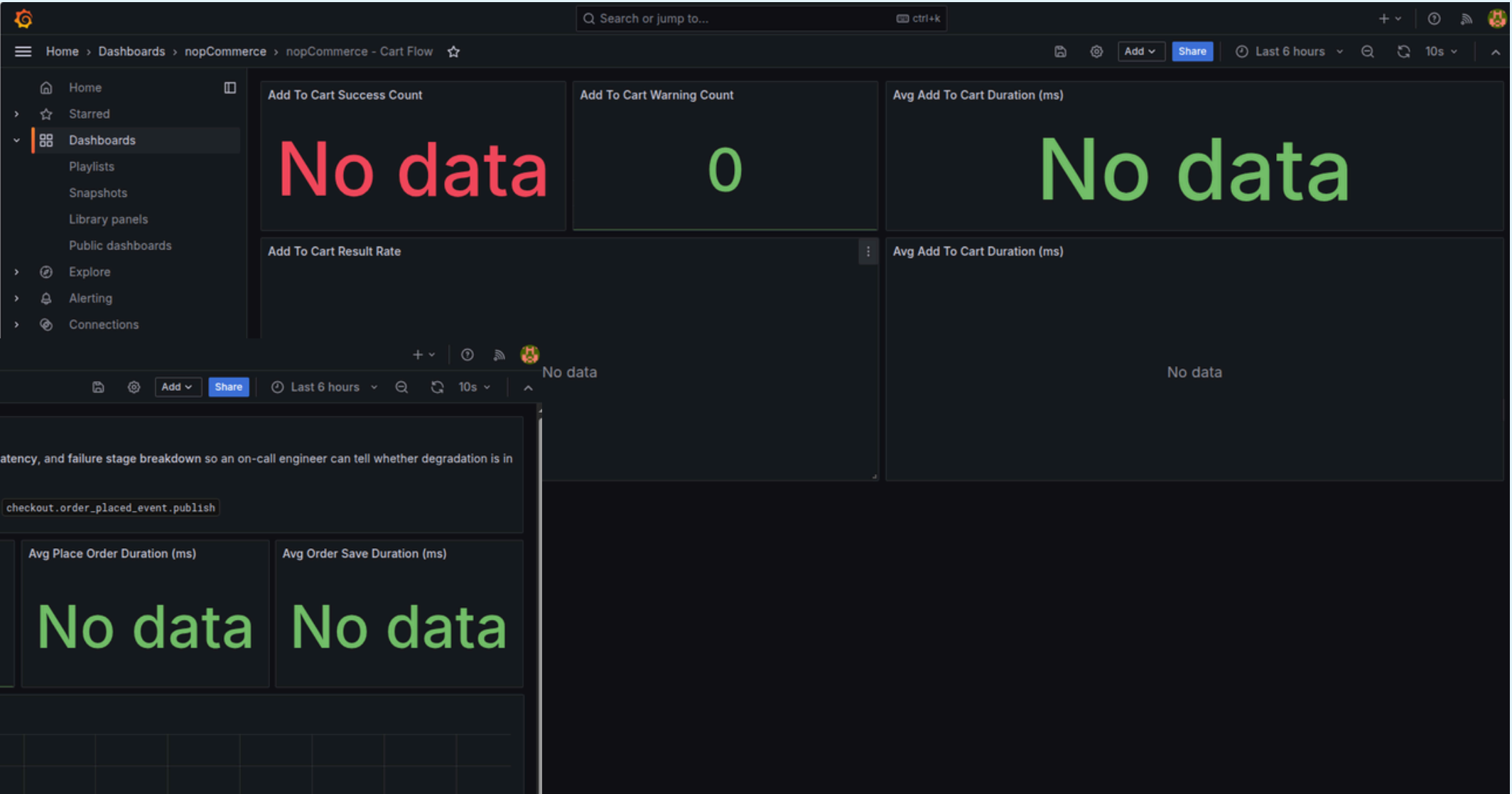
Spans

Span	Wraps	Purpose
checkout.opc_confirm	Full One Page Checkout confirm action	Captures the full controller-layer latency including session validation.
checkout.place_order	Entire order placement logic	Root business span for the entire order placement.
checkout.payment.process	IPaymentPluginManager.ProcessPaymentAsync call	Isolates payment processing latency and outcome from the rest of order placement.
checkout.order.save	EntityRepository<Order>.InsertAsync call	The only non-retryable step in checkout. If this span's duration spikes while checkout.place_order is stable, the overhead is in post-save processing
checkout.order_placed_event.publish	IPublisher.PublishAsync call	Wraps the event fan-out to all OrderPlacedEvent consumers.
cart.add.service	Entire add-to-cart logic	Instrument the add-to-cart operation separately from checkout. Useful for distinguishing cart-layer problems.

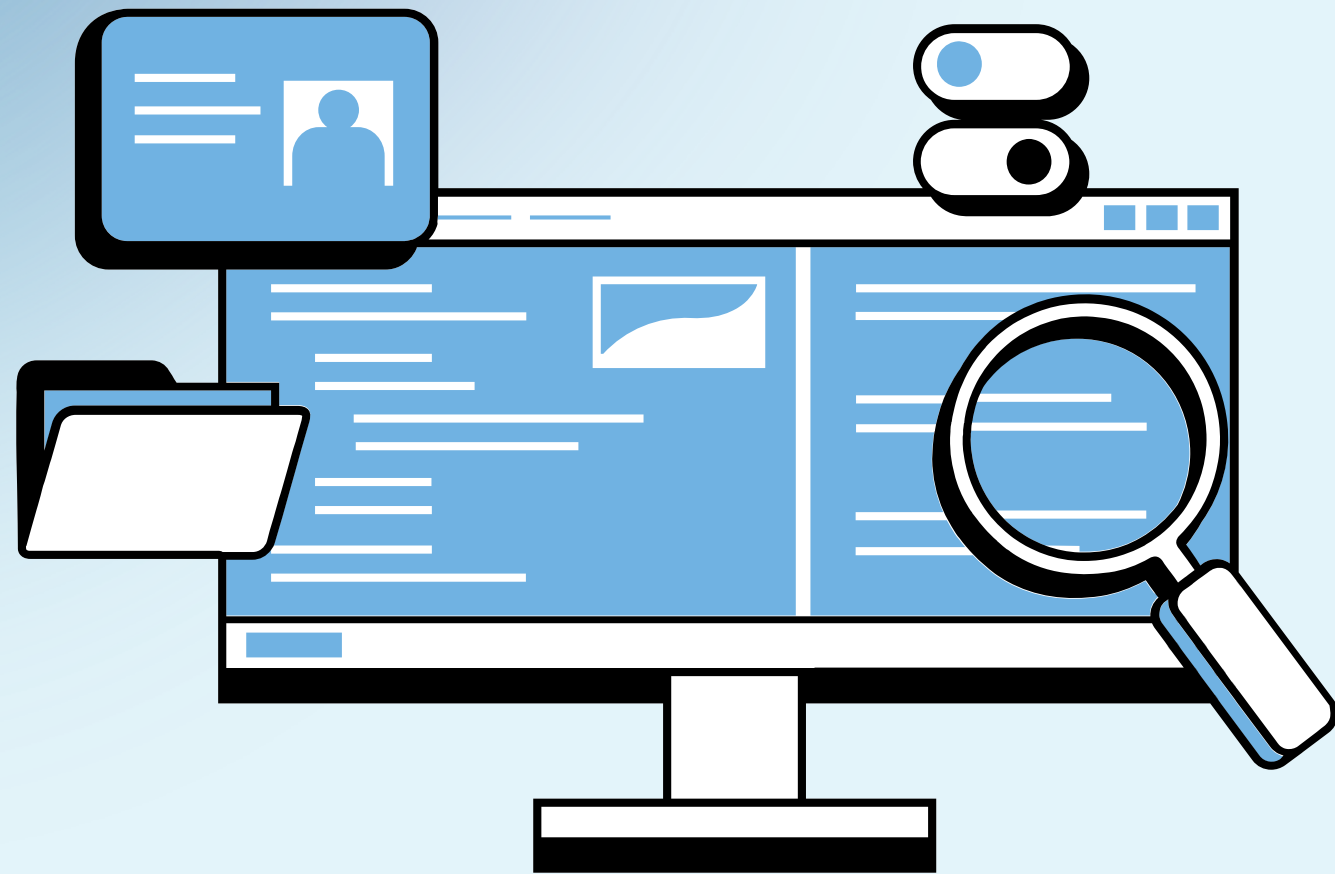
Metrics

Cart Dashboard

Checkout Dashboard



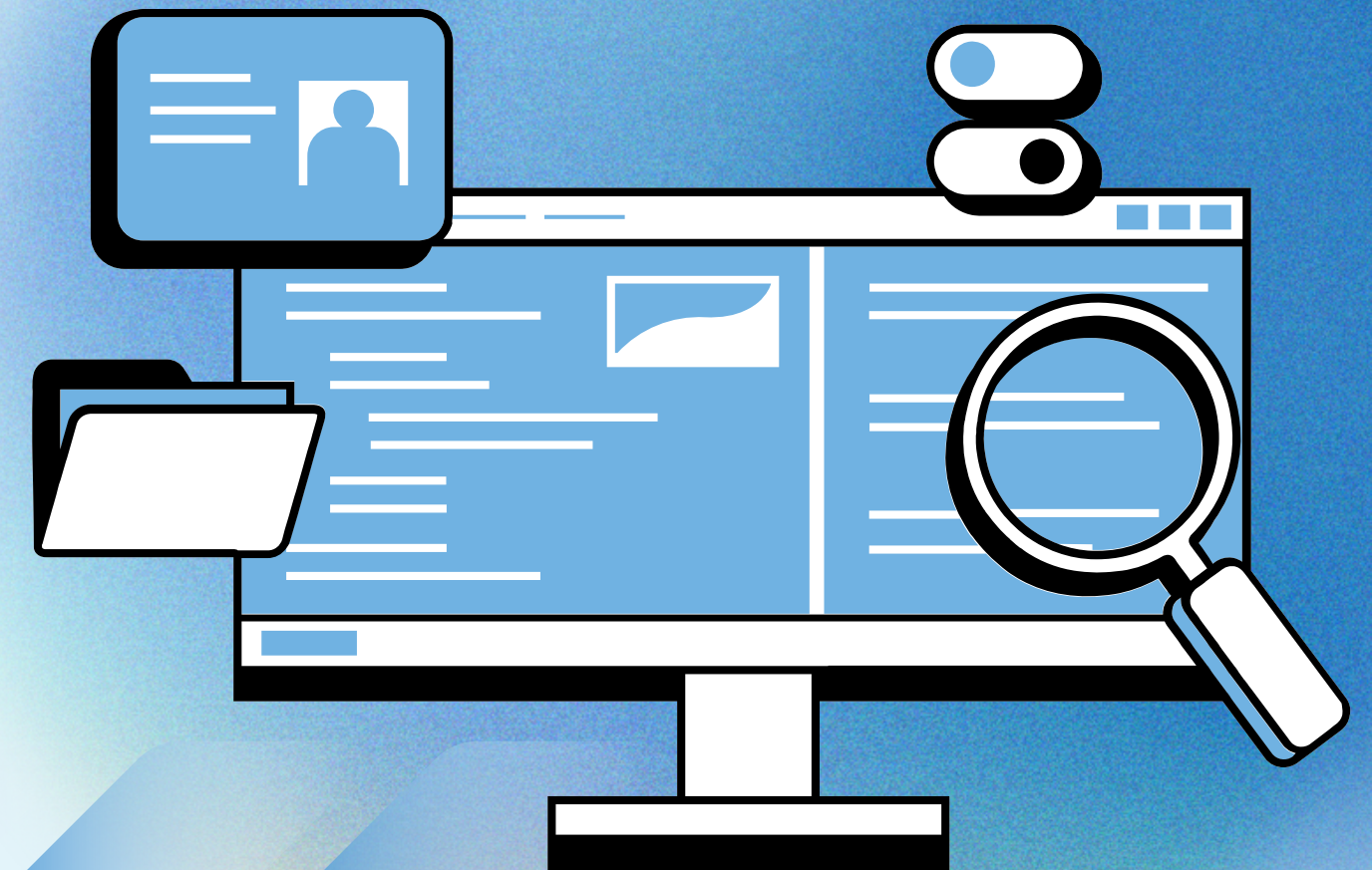
PIIs Strategy



- Never set: customer name, email, billing/shipping address, card number, customer ID, order total.
- These appear in PlaceOrderAsync's parameters and in the Order entity but were deliberately excluded from every SetTag call.
- Sanitization happens at the Otel SDK, in process. TelemetrySanitizer process runs before any export, removing fields with tags mentioned above.

Live Demo

- Simulate system under load:
 - happy path: 10 VUs, full checkout flow (round robin for payment methods)
 - invalid_checkout: 5 VUs, beliberately broken flow (confirms order with missing steps)
- Grafana Dashboards
- Jaeger traces
- Feature Flag to produce system failures



Main trade-offs

- Chose to instrument at decision boundaries like payment and database commit, and left out implementation details of instrumentation. → Less noise, but less precise root cause isolation.
- Remove PII at the SDK → Redaction at the SDK means the application enforces its own privacy boundary. Infrastructure misconfiguration cannot leak PII.
- However it was an heuristic solution, renamed PII tags may pass though.

Thanks

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